

E-Commerce Customer Analytics Report

This analysis evaluates two years of operational data from a Missouri-based e-commerce company to assess key business decisions around staffing, product strategy, and customer incentives. Using **buyer, product, and sales datasets**, the objective is to address five core questions related to order volume trends, profitability, referral program impact, and seasonal performance.

The analysis was conducted in Python (Google Colab) and structured into a series of logical steps covering data preparation, integration, and insight generation. Particular focus was placed on ensuring data quality and aligning outputs with business decision-making.

Given Assumptions

The following assumptions were given:

- The ecommerce company is fully located in the US state of Missouri
- Timestamp of purchase is based on the customer's timezone
- Customers who are referred receive 10% off their first order
- The customers pay shipping fees based on the total cost of their order following the table below

Order Total	Shipping Price
< \$50	\$7.99
\$50 - 99.99	\$4.99
>= \$100	Free (\$0.00)

- The company uses a 3rd party carrier and pays a flat shipping fee of \$4.99 per order, regardless of size or weight.
- The company runs a Black Friday sale (held the Friday after Thanksgiving known for major sales and discounts) on all items purchased that day of 20% reduction (Black Friday hours based on customer's time zone) - this is not reflected in the sales price data
- Ignore any taxes as those are pre-calculated into the data

Analytical Assumptions (applied throughout implementation)

The following assumptions were applied to align with business context and problem constraints:

- Company operates in Missouri → all analysis standardized to CST/CDT
- Timestamps reflect customer-local time → converted per buyer timezone
- Referral discount (10%) applies only to the first order and is considered consumed once used
- Black Friday discounts (20%) applied based on customer-local date
- Discounts are stackable
- Shipping fees are based on post-discount order value
- Cancelled orders are included in volume metrics but excluded from revenue and profit calculations
- Taxes are excluded, as they are pre-accounted for in the dataset

Step 1 – Environment Setup

This step involves importing required libraries and configuring the working environment. No transformations or analytical logic are performed at this stage; it serves solely to establish a consistent and reproducible setup for subsequent analysis.

Step 2 – Data Loading and First Look

The three datasets (*buyer*, *product*, and *sales*) were loaded into Python using pandas from session-based storage. Prior to ingestion, the product data—originally distributed across multiple vendor files (11 datasets)—was consolidated into a single dataset with a consistent schema and standardized boolean fields (treating values such as TRUE/FALSE, Y/N, and 1/0 as binary equivalents), to ensure compatibility during downstream joins.

{Excel formula: =IF (OR (cell=TRUE, cell="TRUE", cell="Y", cell=1), TRUE, FALSE)}

An initial structural review (.info(), .head()) was conducted to understand dataset composition.

Dataset sizes after loading

Dataset	Rows	Columns	Purpose
buyer.xlsx	52,500	14	Customer profiles, segmentation, geography, and referral indicators
product.xlsx	1,037	11	Product catalog, pricing, margins, and vendor attributes
sales.xlsx	376,004	6	Transaction-level order details, timestamps, and fulfillment status

Initial Data Quality Observations

Early inspection identified several issues with potential impact on downstream analysis:

- **Missing join keys:**
 - ~5% of records in the buyer dataset have missing buyer_id, limiting their ability to be joined with transaction data
 - A small number of transactions in the sales dataset also lack buyer_id, resulting in orphan records
- **Time-related inconsistencies:**
 - order_datetime and signup_date are stored as text and require conversion to datetime for time-based analysis
 - Missing timezone values in buyer data may affect accurate time alignment
- **Incomplete fields:**
 - Missing values in quantity, order_status, and order_datetime within the sales dataset
 - Missing region values in buyer data, which may impact geographic segmentation

Step 3 – Data Quality Audit

A systematic data audit was conducted using a reusable *run_quality_checks()* function, which evaluated data completeness, consistency, and validity across all datasets. Each issue was assigned to a severity level based on its impact on downstream analysis:

- **Critical:** Directly compromises metric accuracy or prevents dataset integration
- **Significant:** Introduces bias or affects specific business analyses
- **Minor:** Limited impact; does not materially alter conclusions

Key Data Quality Issues and Resolutions

Critical Issues (Resolved Prior to Analysis)

- **Duplicate transactions:** ~35% of sales records (131K rows) were exact duplicates. These were removed to prevent significant overstatement of order volume and revenue.
- **Invalid timestamps (1970):** ~4.8K records contained default Unix epoch dates, indicating corrupt entries; these were excluded.
- **Product margin inconsistencies:** Invalid profit margins (negative or >100%) were corrected through normalization and validation checks.
- **Missing join keys:** Buyer records without buyer_id were removed, as they could not be linked to transactions.
- **Unmatched buyer records:** ~13K sales records lacked corresponding buyer profiles; an inner join was applied, and the resulting data loss was documented as a limitation.

Significant Issues (Addressed to Improve Analytical Accuracy)

- **Datetime inconsistencies:** order_datetime and signup_date were converted to proper datetime formats to enable all time-based analyses.
- **Missing timezone data (~5%):** Imputed using state-level mapping to ensure accurate hourly trend analysis.
- **Missing geographic data (~10% region):** Retained with fallback to state-level analysis where needed.
- **Quantity anomalies:** Extreme outliers (e.g., unrealistically large order quantities) were capped to prevent distortion in aggregated metrics.
- **Invalid categorical values:** Non-standard values (e.g., is_referred = 2) were treated as missing and excluded from relevant analyses.
- **Incomplete transaction fields:** Records missing critical fields such as order_datetime or order_status were removed.

Minor Issues (Corrected for Consistency)

- Standardized categorical formatting (e.g., typos, casing inconsistencies)
- Converted data types (e.g., quantity from float to integer)
- Resolved encoding issues in text fields
- Consolidated duplicate product entries across vendors using active vendor and pricing logic

Data Reduction and Impact

Following data cleaning, the sales dataset was reduced from ~376K to ~238K records. While initially significant (~37%), most of this reduction (~95%) was due to removal of duplicate transactions, indicating that the raw dataset substantially overstated actual order volume.

After deduplication, the effective transaction base was approximately ~244K records, with subsequent reductions driven by necessary data quality filters and join constraints.

Step 4 – Cleaning and Feature Engineering

This step operates the data quality decisions identified earlier and prepares the dataset for analysis. Cleaning and feature engineering were performed on copied datasets to preserve raw data integrity.

Part A – Cleaning

Data cleaning was applied in a structured sequence to ensure dependencies were handled correctly (e.g., deduplication prior to filtering). Key transformations included:

- **Deduplication and filtering:**
Removed duplicate transactions and excluded records with missing critical fields (e.g., buyer_id, order_datetime, order_status, quantity) to ensure analytical consistency.
- **Datetime standardization:**
Converted order_datetime and signup_date to datetime format and removed invalid timestamps (e.g., default Unix epoch values).
- **Product data corrections:**
Resolved inconsistencies in profit_margin caused by mixed formats (decimal vs percentage) across vendor sources. Values were normalized to a consistent scale and validated for accuracy.
- **Categorical and encoding fixes:**
Standardized categorical fields (e.g. casing inconsistencies, status typos) and corrected encoding issues in text fields.
- **Missing data handling:**
Imputed missing timezone values using state-level mapping and retained incomplete geographic fields with documented fallback logic.
- **Outlier treatment:**
Extreme quantity values were capped to prevent distortion in aggregated metrics.

Key Data Resolution Decisions

- **Profit Margin Normalization:**

The profit_margin field contained a mix of decimal (0–1) and percentage (0–100) formats due to inconsistent source schemas. Values above 1 were scaled appropriately, while negative margins were retained where they plausibly reflected real business scenarios (e.g., loss-leading products).

- **Duplicate Product Entries (Multi-Vendor):**

Some products were associated with multiple vendors and prices, while transaction data did not specify vendors. To avoid duplication during join:

- Active vendors were prioritized
- Where multiple active vendors existed, the lower price was selected to maintain conservative revenue estimates

This ensured a one-to-one product mapping and prevented artificial inflation of transaction records during dataset join.

Part B – Feature Engineering

New fields were created to directly support the business questions, grouped into three categories:

Temporal Fields

- Standardized timestamps to company time (CST/CDT) using buyer-level timezone conversion
- Derived hourly, monthly, and quarterly fields to support time-based analyses (e.g., staffing evaluation, seasonal trends)

Flag Fields

- Identified **Black Friday orders** based on customer-local time
- Flagged **first referred orders** to isolate referral program impact
- Created a **revenue eligibility flag** to exclude non-delivered orders from financial metrics

Financial Fields

- Calculated order-level metrics which include:
 - Subtotal, discounts (referral and Black Friday), and final transaction value
 - Shipping fees based on pricing tiers
 - Carrier costs per delivered order
 - Net profit, incorporating margins, discounts, and fulfillment costs

These features enabled consistent and comparable measurements of revenue, profitability, and promotional impact across all analyses.

Step 5 – Join Function

To support scalable and reusable data integration, a parameterized join function was implemented to dynamically combine sales and buyer data based on analysis needs. Rather than creating a single, fully merged dataset, the function allows selective inclusion of buyer attributes, reducing unnecessary data overhead and improving flexibility across different use cases.

Join Strategy

- **Sales → Buyer (INNER JOIN on buyer_id):**

An inner join was used to ensure all retained records have valid buyer information. This resulted in the exclusion of ~22.9K transactions without matching buyer profiles, preventing incomplete segmentation in downstream analyses.

- **Sales → Product (LEFT JOIN on composite key):**

Product attributes were merged earlier using product_num + subcategory_id to uniquely identify products and avoid duplication caused by multi-vendor entries.

The final analysis dataset contains approximately **215K records and 41 columns**, representing a clean and fully integrated view of transactions.

Step 6 – Answering Business Questions

Q1. What is order volume by hour (based on company time zone)? Does this order volume support the need to maintain operational staffing of 24/7 for customer fulfillment.

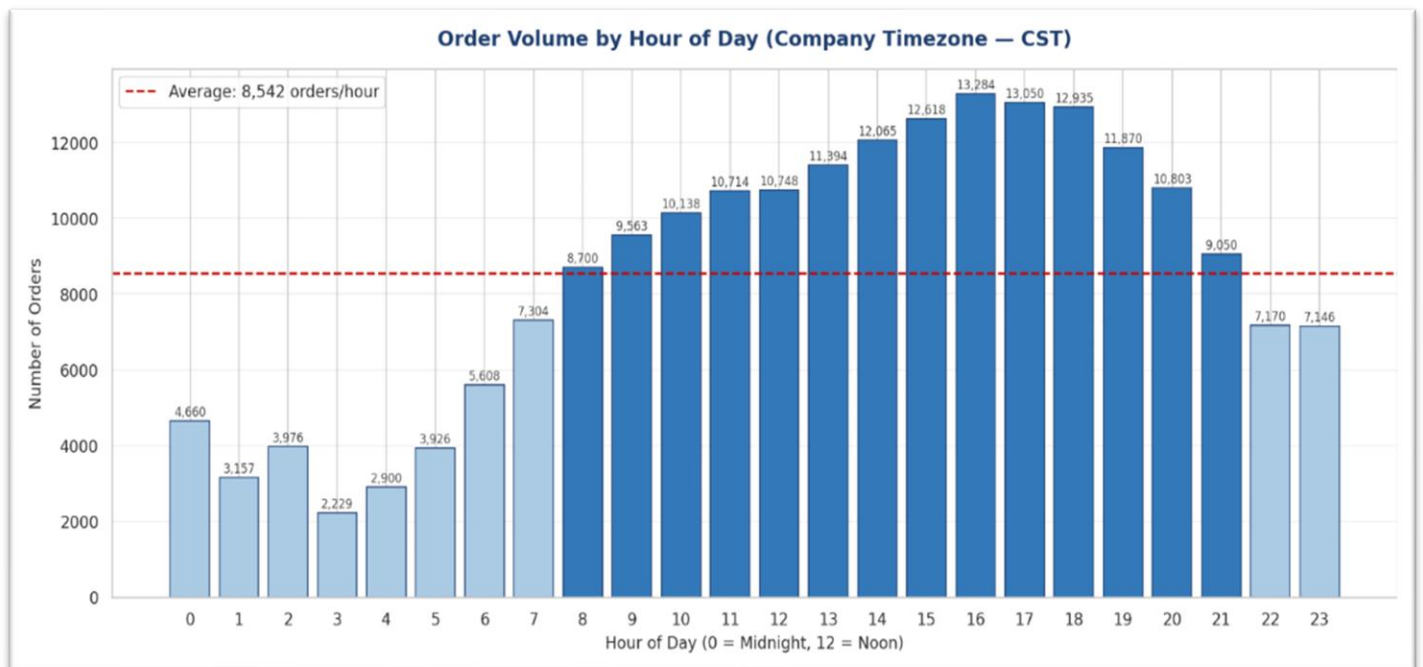
Approach

Order volume was analyzed by hour using timestamps standardized to CST (company timezone). A total of ~205K orders were evaluated to identify demand patterns across the day.

Key Insights

Order activity follows a clear and consistent daily pattern:

- **Peak demand occurs in the afternoon and early evening (12 PM – 8 PM CST):** ~46% of all orders are concentrated in this 8-hour window, with the highest volume at 4 PM (~6.5% of daily orders).
- **Moderate demand during morning and late evening (8 AM – 12 PM, 8 PM – 11 PM):** Each hour contributes ~3.5%–5% of daily volume, indicating steady but lower activity than peak hours.
- **Significantly reduced overnight activity (12 AM – 7 AM):** Only ~17% of total orders occur during this entire 8-hour window, with the lowest point at 3 AM (~1% of daily volume).
- **Demand imbalance is substantial:** The peak hour processes ~6× more orders than the lowest hour, highlighting a strong mismatch in workload across the day.



Business Interpretation

The distribution of order volume does not support a uniform 24/7 staffing model. Demand is highly concentrated during specific hours, while overnight activity remains consistently low.

Maintaining equal staffing levels across all hours would likely result in underutilized labor during low-demand periods and inefficient cost allocation. However, the presence of steady (though low) overnight demand indicates that a complete shutdown is not feasible.

Recommendation

Transition from a flat 24/7 staffing model to a tiered, demand-aligned staffing strategy:

- **Full staffing (12 PM – 8 PM CST):** Align capacity with peak demand periods
- **Moderate staffing (8 AM – 12 PM, 8 PM – 11 PM):** Support steady but lower volume
- **Minimal staffing (12 AM – 7 AM):** Maintain a lean team to handle overnight orders

This approach balances cost efficiency with operational continuity, ensuring resources are allocated where they generate the most value.

Success Metrics

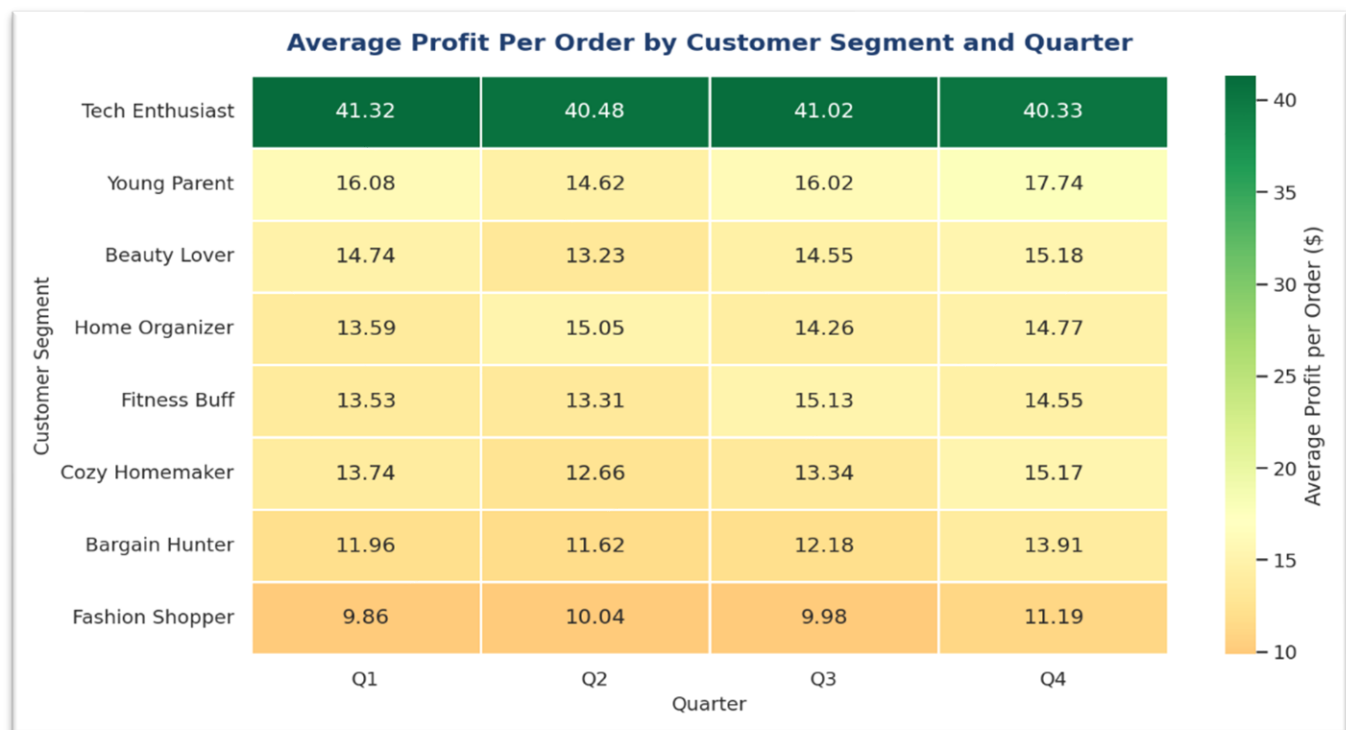
To evaluate the effectiveness of this change, the company should monitor:

- **Order-to-ship time (by hour):** Ensure fulfillment delays do not increase overnight
- **On-time delivery rate:** Maintain adherence to promised delivery timelines
- **Labor cost per order (by shift):** Measure efficiency gains from staffing adjustments
- **Overnight backlog volume:** Ensure order accumulation remains controlled
- **Customer feedback on fulfillment speed:** Detect early signs of service degradation

Q2. What is the average profit per order by quarter for each customer segment?

Approach

Average net profit per order was calculated for each customer segment across quarters, considering only delivered orders to ensure accurate profitability measurement.



Key Insights

- **Significant variation across customer segments:** Profitability differs substantially by segment, with **Tech Enthusiasts (~\$40.8 per order)** generating 2.5–4× higher profit than all other segments.
- **Low-performing segments:** Fashion Shoppers (~\$10.3 per order) consistently generate the lowest profit, less than one-fourth of Tech Enthusiasts.
- **Limited seasonal variation:** Profit per order remains relatively stable across quarters within each segment, indicating that customer segment is a stronger driver of profitability than seasonality.
- **Moderate Q4 uplift:** Most segments show a slight increase in Q4, likely driven by holiday demand, though the impact is not large enough to alter overall segment rankings.

Business Interpretation

Customer segment is the primary driver of profitability, not timing. The large gap between segments indicates that **all customers are not equally valuable**, even if order volume is similar.

High-performing segments (e.g., Tech Enthusiasts) likely purchase higher-value products or bundles, while lower-performing segments (e.g., Fashion Shoppers) generate smaller margins per transaction. This suggests that growth strategies focused purely on increasing order volume may not maximize profitability unless segment mix is considered.

Recommendation

- **Prioritize acquisition of high-value segments:** Increase marketing investment in channels that attract high-profit segments such as Tech Enthusiasts.
- **Improve profitability of low-value segments:** For segments like Fashion Shoppers and Bargain Hunters, focus on increasing average order value through bundling, cross-selling, or premium product offerings.
- **Leverage seasonal trends selectively:** Target segments that show Q4 responsiveness with seasonal campaigns, while maintaining consistent engagement for segments with stable year-round behavior.
- **Replicate high-value behavior:** Analyze purchasing patterns of high-profit segments and apply similar product or pricing strategies to other segments where feasible.

Success Metrics

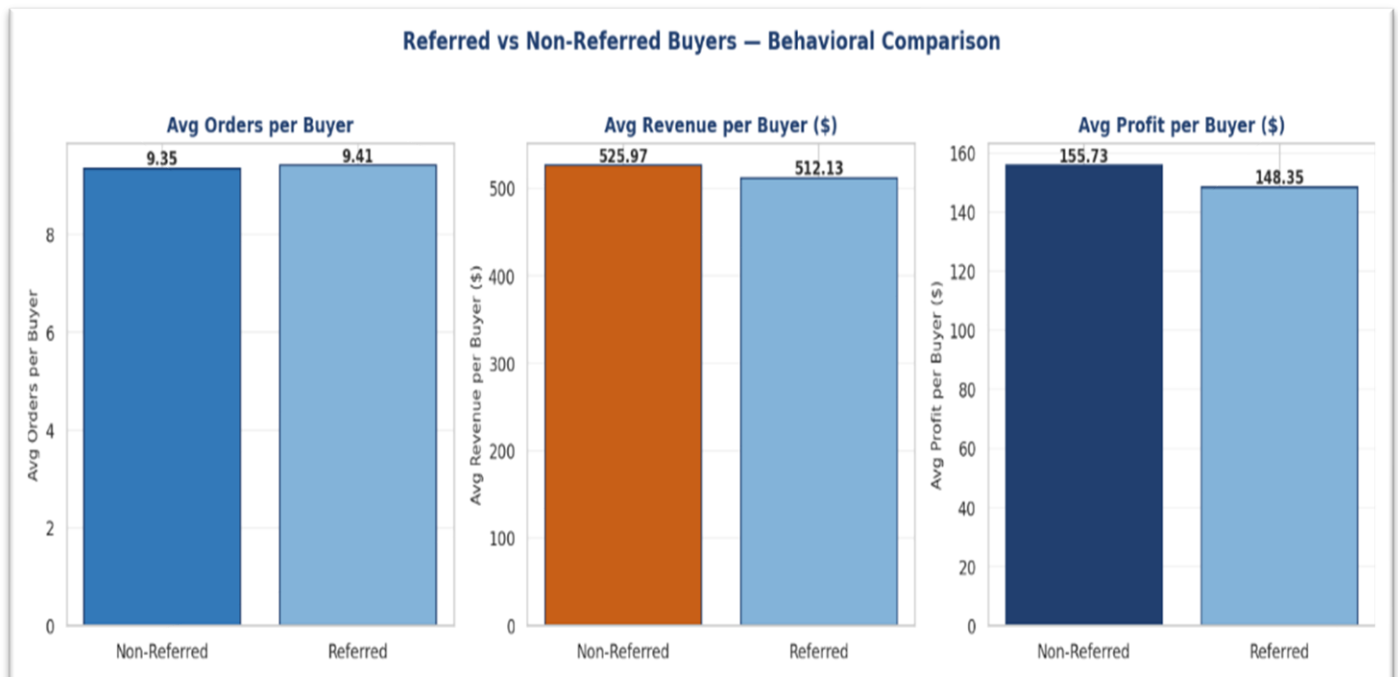
To evaluate the effectiveness of these changes, the company should monitor:

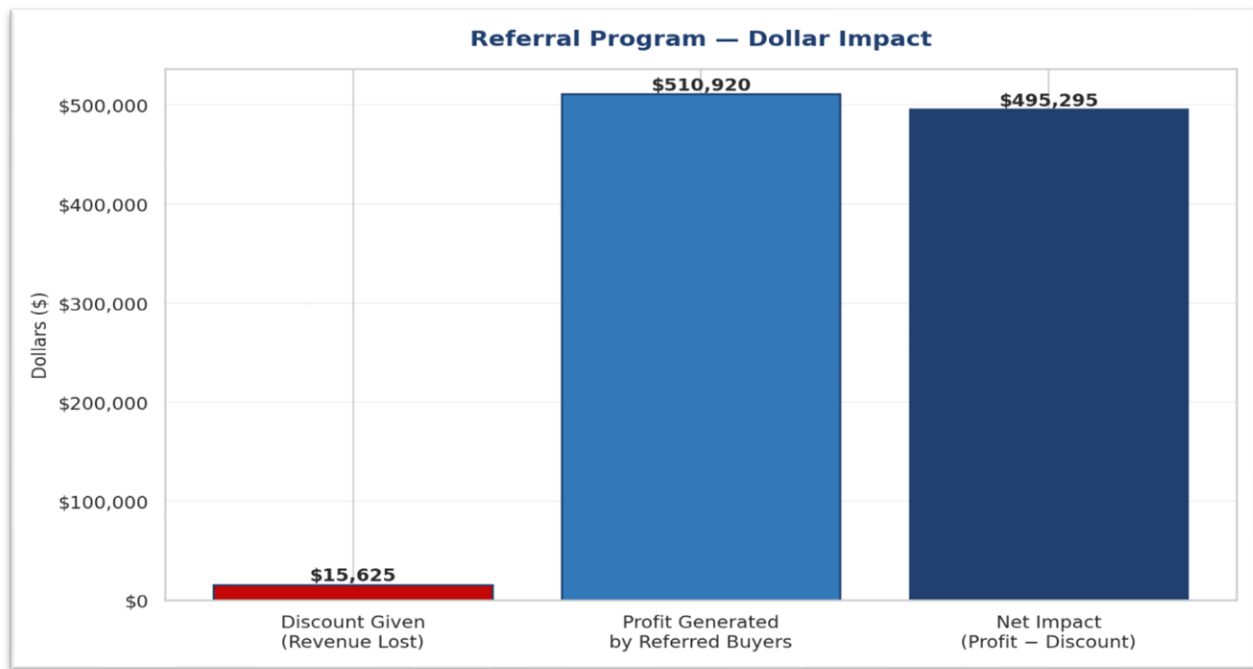
- **Profit per order (by segment):** Ensure targeted segments (e.g., lower-performing ones) show improvement over time
- **Segment mix of new customer acquisition:** Track whether higher-value segments (e.g., Tech Enthusiasts) are increasing as a share of new customers
- **Average order value (AOV) by segment:** Validate whether cross-sell and bundling strategies are increasing spend in lower-profit segments
- **Customer lifetime value (CLV) by segment:** Measure long-term profitability, especially for segments with lower per-order profit but higher repeat behavior
- **Cost per acquisition (CPA) vs. profit per order:** Ensure marketing spend remains efficient relative to segment-level profitability

Q3. What is the total amount of revenue lost due to our referral program? Should we continue to offer discounts for referred Customers?

Approach

The financial impact of the referral program was evaluated by quantifying total discounts applied to first-time referred orders and comparing the behavior and profitability of referred vs. non-referred buyers.





Key Insights

- **Low cost of discounts:** Total revenue foregone from the referral program is **~\$15.6K across ~3.6K orders**, with an average discount of **\$4.35 per order**.
- **Referred buyers perform similarly to non-referred buyers:**
 - Average orders per buyer: ~9.4 (referred) vs ~9.3 (non-referred)
 - Average profit per buyer: ~\$148 vs ~\$156 (This indicates that referred customers are **not lower-quality or one-time buyers**)
- **Strong overall profitability:** Referred buyers generated **~\$511K in total profit**, resulting in **~\$495K net profit after accounting for discount costs**.
- **Exceptional return on investment:** The program delivers approximately **\$32.7 in profit for every \$1 of discount**, indicating a highly efficient acquisition channel.

Business Interpretation

The referral program is highly cost-effective. The discount acts as a **low-cost acquisition incentive**, while referred customers exhibit behavior comparable to organically acquired users.

The minimal gap in profit per buyer suggests that the discount primarily impacts on the first transaction, without reducing long-term customer value. Compared to typical paid acquisition channels, the referral program delivers **substantially higher returns at a fraction of the cost**.

Recommendation

- **Continue the referral program:** Given its strong ROI, the program should be maintained as a core acquisition strategy.
- **Expand and optimize the program:**
 - Test higher or alternative incentives to increase referral volume
 - Introducing rewards for referrers to drive participation
 - Target high-value segments (e.g., Tech Enthusiasts) to amplify impact
- **Position as a primary growth lever:** Given its efficiency, referral-based acquisition should be prioritized relative to more expensive marketing channels.

Success Metrics

To evaluate the ongoing effectiveness of the referral program, the company should monitor:

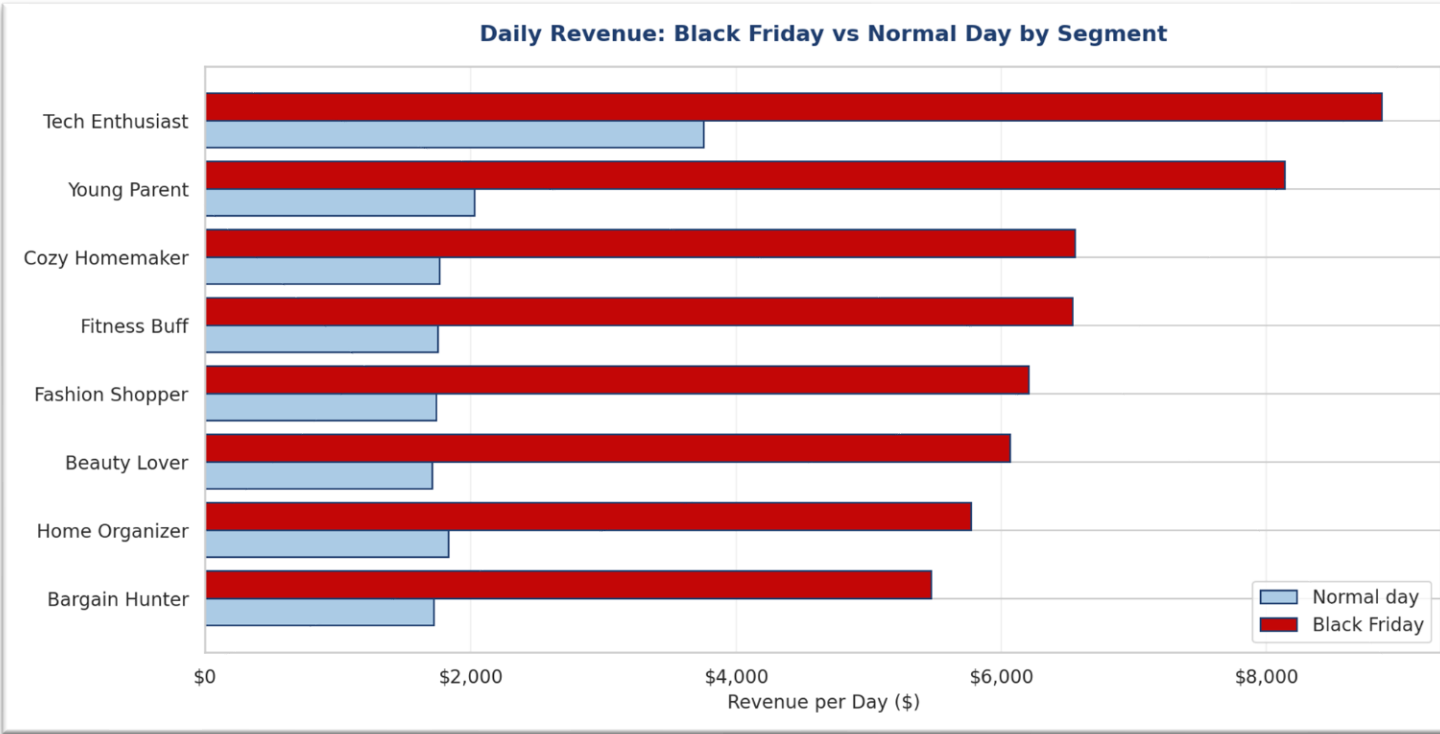
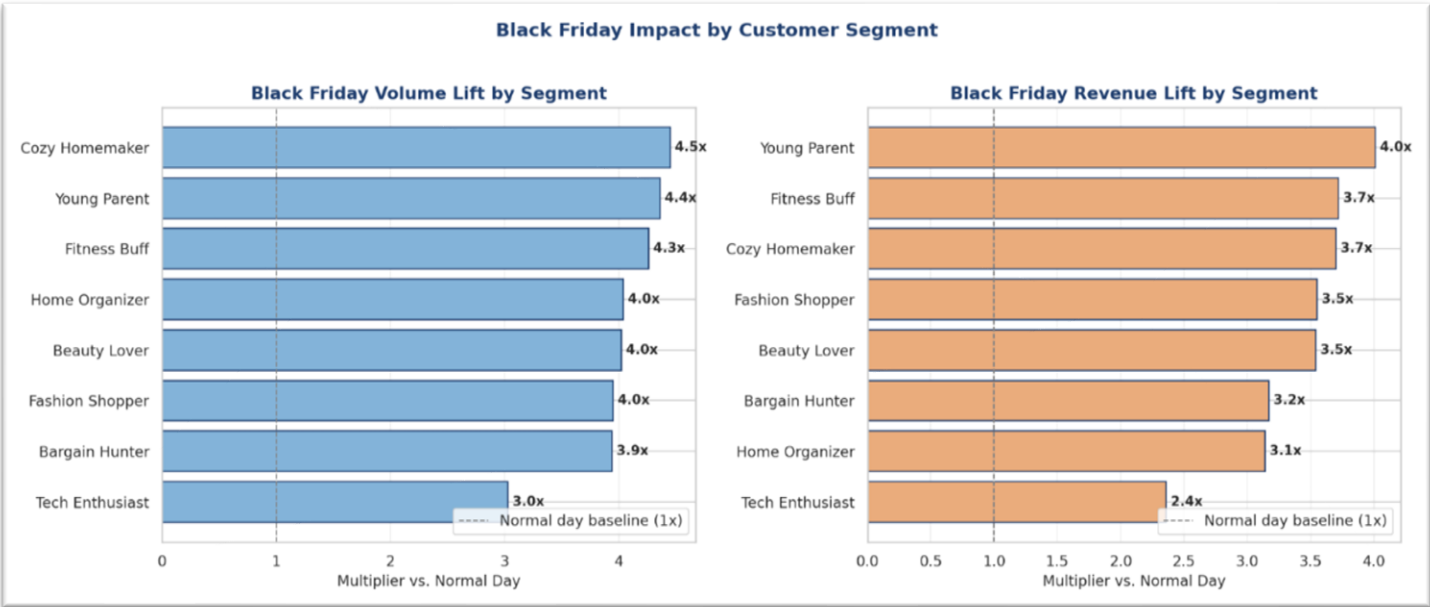
- **Return per dollar discount:** Ensure the program continues to deliver strong ROI over time
- **Referred buyer retention rate:** Measure the % of referred buyers placing repeat orders

- **Average orders per referred buyer:** Track long-term engagement vs. non-referred buyers
- **Segment mix of referred customers:** Ensure acquisition skews toward higher-value segments
- **Referral participation rate:** % of existing customers generating referrals
- **Cost per acquisition (CPA) via referrals vs. other channels:** Benchmark efficiency against alternative marketing expenses.

Q4. What is the impact of Black Friday on Annual Sales (in terms of both volume and revenue) by customer segment?

Approach

Black Friday performance (two days across two years) was compared against average daily performance to measure impact on order volume and revenue. Segment-level analysis was conducted to assess differences in promotional responsiveness.



Key Insights

- **Significant short-term lift:**
 - Order volume increases **~4.0×** (1,166 vs 291 daily orders)
 - Revenue increases **~3.3×** (\$53.6K vs \$16.3K daily)

This confirms Black Friday as a high-impact promotional event.
- **Limited annual contribution:**
 - **~1.1% of annual order volume**
 - **~0.9% of annual revenue**

Despite strong daily performance, Black Friday has a **minimal impact on overall yearly results**.
- **Discount-driven dynamics:** Volume growth exceeds revenue growth due to the 20% discount, indicating increased purchasing at lower per-order value.
- **Segment-level variation in response:**
 - **Highly responsive segments:** Cozy Homemaker, Young Parent, Fitness Buff (~4.2×–4.5× volume lift)
 - **Low responsiveness:** Tech Enthusiasts (~3.0× volume, ~2.4× revenue lift)
 - **Top revenue contributors:** Tech Enthusiasts and Young Parents, driven by higher average order values

Business Interpretation

Black Friday is effective at driving short-term demand but is **not a primary driver of annual performance**. The event generates incremental volume, but its overall contribution remains small relative to total yearly sales.

Customer segments respond differently to promotions. Price-sensitive segments significantly increase purchasing during Black Friday, while high-value segments (e.g., Tech Enthusiasts) maintain relatively stable behavior, suggesting **lower** price sensitivity and stronger baseline demand.

The gap between volume and revenue lift indicates that discounts play a major role in driving activity, raising questions about profit efficiency during the event.

Recommendation

- **Continue Black Friday but optimize discount strategy:** Replace the flat 20% discount with a more targeted or tiered approach to better aligned discounts with product margins and segment behavior.
- **Target promotion-sensitive segments:** Focus marketing efforts on segments that show strong lift (e.g., Cozy Homemaker, Young Parent), where discounts meaningfully change behavior.
- **Avoid over-reliance on Black Friday:** Given its low contribution to annual revenue, prioritize consistent year-round performance rather than relying on a single promotional event.
- **Leverage high-value segments differently:** Maintain engagement with segments like Tech Enthusiasts without heavy discounting, as their purchasing behavior is less promotion driven.
- **Explore extended promotional windows:** Expanding beyond a single day (e.g., multi-day events) may capture additional demand without concentrating activity into a narrow timeframe.

Success Metrics

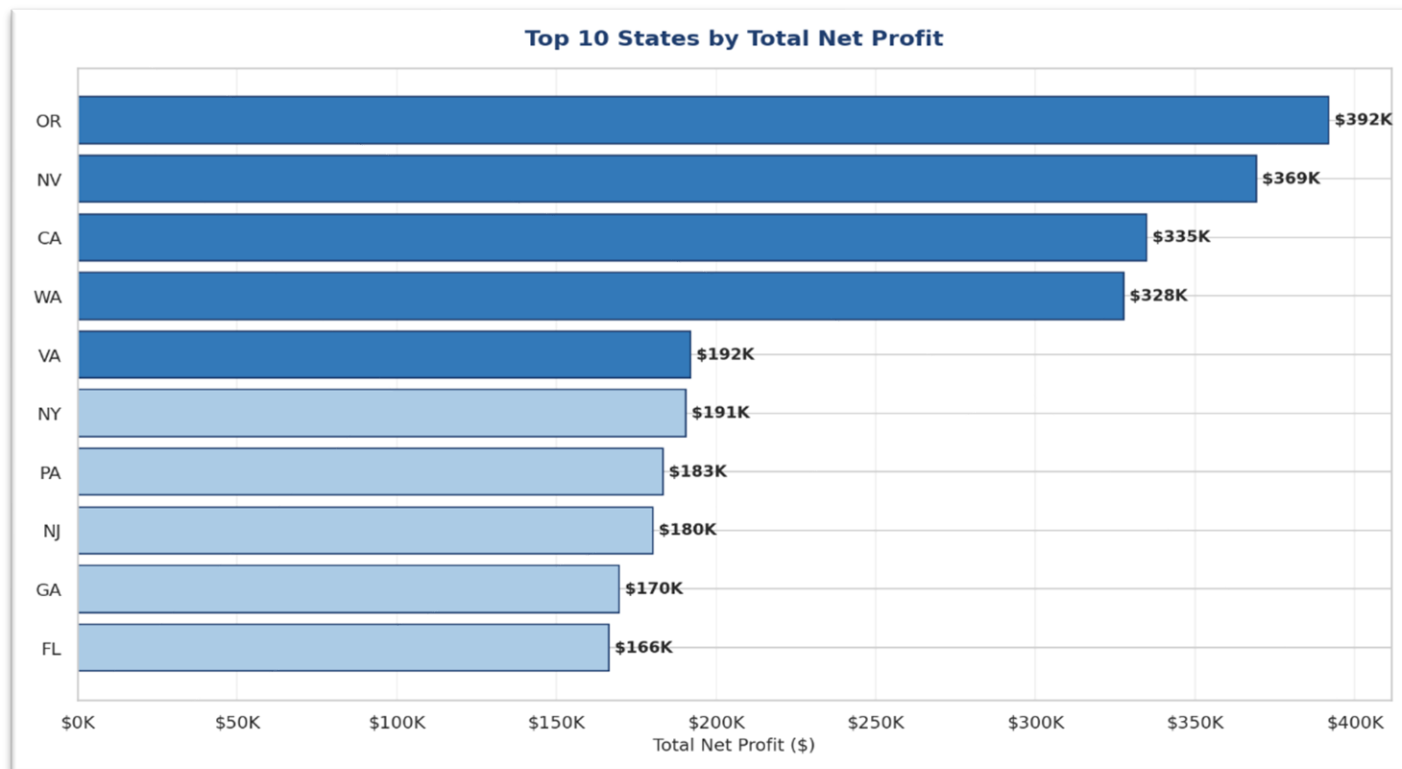
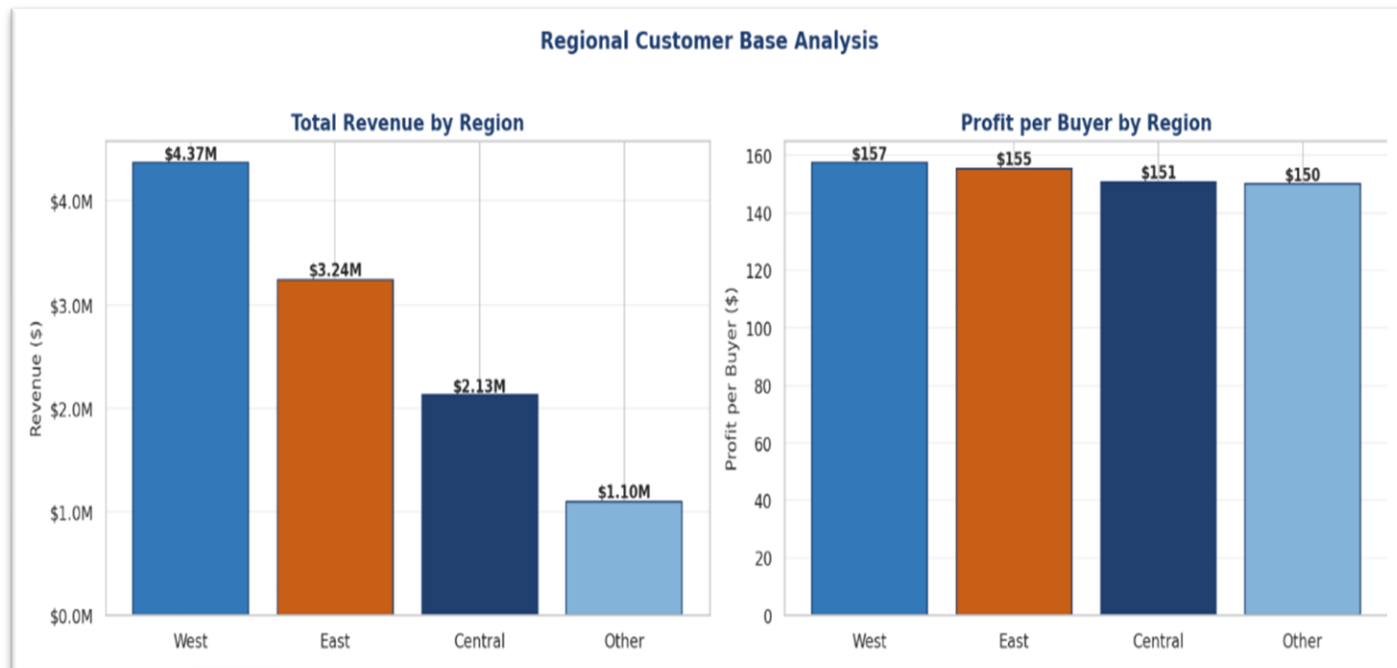
To evaluate the effectiveness of Black Friday strategy, the company should monitor:

- **Black Friday revenue as % of annual revenue:** Track dependence on the event over time
- **Discount ROI (incremental revenue vs discount cost):** Ensure promotions generate net value
- **Net profit during Black Friday vs normal days:** Validate that higher volume translates into profitability
- **Segment-level lift (volume and revenue):** Track changes in promotional responsiveness
- **New vs returning customer mix:** Assess whether Black Friday drives acquisition or shifts existing demand
- **Cart abandonment rate during Black Friday:** Identify potential operational or capacity constraints

Q5. Future Next Steps: Where is our best customer base and what are some KPIs the company should track to acquire more customers here?

Approach

Geographic performance was evaluated across regions and states using order volume, revenue, and profit metrics. Segment-level contribution within each geography was analyzed to identify the most valuable customer profiles.



Key Insights

- **West region has the strongest market:** Leads in total orders, revenue, and profit, contributing the largest share of overall performance.
- **Growth driven by volume, not higher-value customers:** Profit per buyer is relatively consistent across regions (~\$150–\$157), indicating that the West's advantage comes from **higher customer concentration**, not superior per-customer value.

- **Top-performing states concentrated on the West Coast:** Oregon, Nevada, California, and Washington account for a significant share of total profit, with Oregon standing out in both total profit and profit per buyer.
- **Consistent segment dominance:** Tech Enthusiasts are the top-performing segment across all regions, contributing the largest share of profit regardless of geography.

Business Interpretation

The company's strongest customer base is geographically concentrated in the **West Coast**, particularly in high-performing states such as Oregon and Nevada. However, customer value remains consistent across regions, suggesting that growth opportunities lie in expanding customer acquisition in high-density markets rather than relying on higher per-customer profitability.

The combination of high-performing geography (West) and high-value segment (Tech Enthusiasts) represents the most attractive growth opportunity.

Recommendation

- **Focus geographic expansion on the West Coast:** Prioritize acquisition efforts in Oregon, Nevada, California, and Washington, where existing demand is strongest.
- **Target high-value segments across all regions:** Continue prioritizing Tech Enthusiasts, as they consistently drive the highest profitability regardless of location.
- **Adopt a more focused growth strategy:** Shift from a broad, "sell to everyone" approach toward targeted acquisition in high-performing regions and segments.
- **Maintain secondary market investment:** Continue supporting strong secondary markets (e.g., East region, high-performing states like New York) to sustain diversified growth.

Success Metrics

To evaluate the effectiveness of this growth strategy, the company should monitor:

- **New customer acquisition rate (by region/state):** Track growth in target markets (e.g., West Coast states)
- **Segment mix of new customers:** Measure share of high-value segments (e.g., Tech Enthusiasts)
- **Customer lifetime value (CLV) by region and segment:** Validate long-term profitability of target customers
- **Customer acquisition cost (CAC) by region:** Ensure efficient spending in priority markets
- **CLV-to-CAC ratio:** Confirm that acquisition investments generate strong returns
- **Repeat purchase rate (e.g., within 90 days):** Measure early customer retention
- **Revenue concentration by region:** Track whether growth is successfully driven by target geographies
- **Average order value (AOV) by region/state:** Monitor changes in purchasing behavior